

Technical Whitepaper - CNC Tap Testing

Low-Cost CNC Tool Chatter Detection and Resonance Analysis Using Raspberry Pi and MPU-6050 Accelerometer Sensors

Author: Robert Zega

Organization: Miyama USA, Inc, 7081 International Dr, Louisville, Ky 40258

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Abstract

Machine tool chatter remains one of the largest barriers to maximizing productivity, surface finish quality, tool life, and process stability in CNC machining operations. Commercial modal testing systems capable of identifying resonance frequencies and generating spindle-speed recommendations often cost several thousand dollars, placing them out of reach for many small manufacturers, makers, and educational institutions.

This paper describes the design and implementation of a low-cost CNC tool resonance analysis system built using a Raspberry Pi 4 and MPU-6050 three-axis accelerometer. The system performs impact (tap) testing, frequency-domain analysis using Fast Fourier Transforms (FFT), damping estimation through logarithmic decrement and half-power bandwidth methods, and automatic generation of spindle-speed avoidance charts.

The resulting platform provides manufacturing engineers with practical resonance information that can be used to reduce chatter and improve machining performance while maintaining open-source accessibility and low implementation cost.

1. Introduction

In CNC machining operations, chatter is a self-excited vibration phenomenon that can significantly impact:

- Surface finish quality
- Tool life
- Material removal rates
- Machine reliability
- Production efficiency

Many machining processes rely on trial-and-error adjustment of spindle speeds to avoid chatter.

Industrial modal testing systems solve this problem by identifying the natural frequencies of the cutting system and allowing operators to avoid operating conditions that excite these frequencies.

However, commercial systems frequently cost thousands of dollars and may not be economically justified for:

- Small machine shops
- Makers
- Technical schools

- Manufacturing engineers performing occasional setup optimization

The objective of this project was to develop an affordable alternative using commonly available Raspberry Pi hardware and open-source software.

2. Problem Statement

Every machining system exhibits natural vibration modes.

These modes are influenced by:

- Tool length
- Tool diameter
- Tool material
- Holder design
- Spindle stiffness
- Fixture stiffness
- Workpiece geometry

When spindle speed causes the tooth-passing frequency to coincide with a structural resonance, the system may enter a chatter condition.

Consequences include:

- Poor surface finish
- Reduced dimensional accuracy
- Increased tool wear
- Tool breakage
- Increased machine downtime

A method was needed to quickly characterize the dynamic behavior of a machining setup before production machining begins.

3. Project Objectives

The project was designed with the following goals:

Primary Objectives

1. Measure vibration response using low-cost hardware
2. Identify dominant resonance frequencies
3. Estimate damping ratio
4. Generate spindle speed recommendations
5. Produce graphical outputs automatically

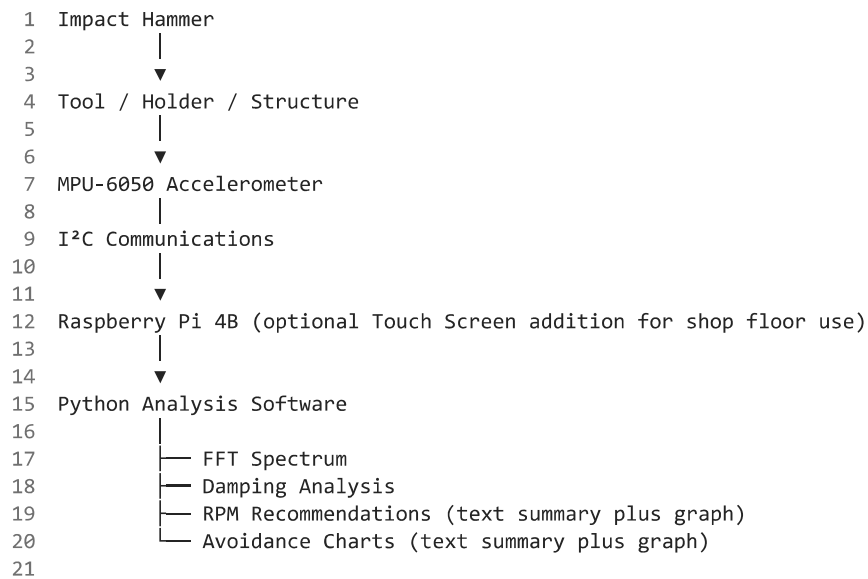
Secondary Objectives

1. Minimize hardware cost
2. Operate without an internet connection
3. Support headless Raspberry Pi deployments

4. Remain fully open source
 5. Be suitable for educational use
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4. System Architecture

The system consists of:



5. Hardware Design

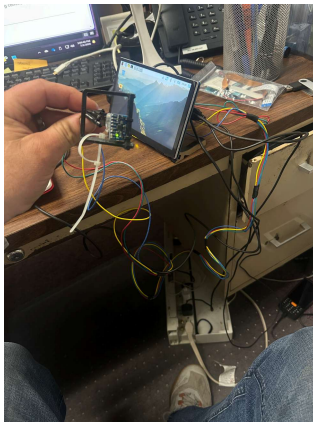
5.1 Raspberry Pi 4B

The Raspberry Pi serves as:

- Data acquisition platform
- Signal processing engine
- User interface
- Reporting platform
- Breakout Board

Advantages include:

- Low cost
- Extensive Python support
- GPIO and I²C capabilities
- Large community support



5.2 MPU-6050 Accelerometer

The MPU-6050 provides:

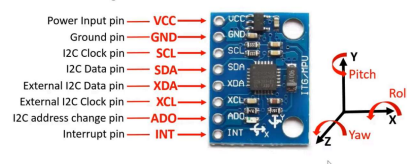
- 3-axis acceleration measurement (x,y,z)
- Digital I²C communication
- Compact form factor
- Low cost

The sensor is mounted to:

- Tool holder
- Spindle housing
- Fixture
- Workpiece

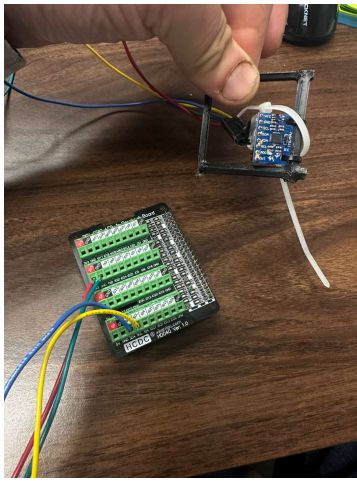
depending on the measurement objective.

Sensor pins & directions



MPU6050

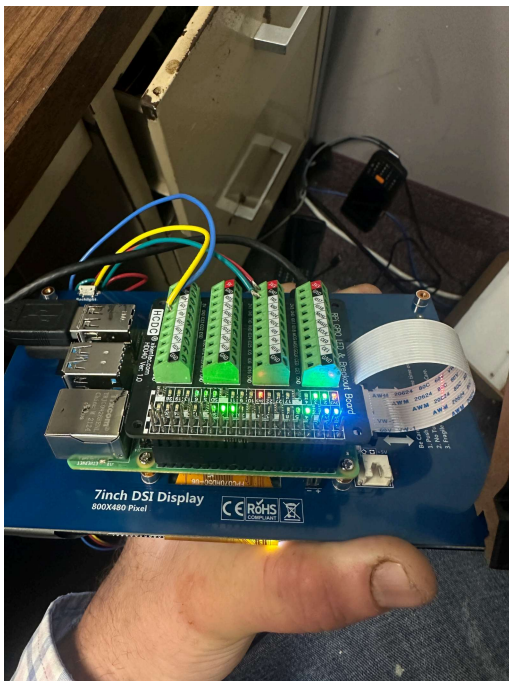
MPU6050 with Arduino - GY-521

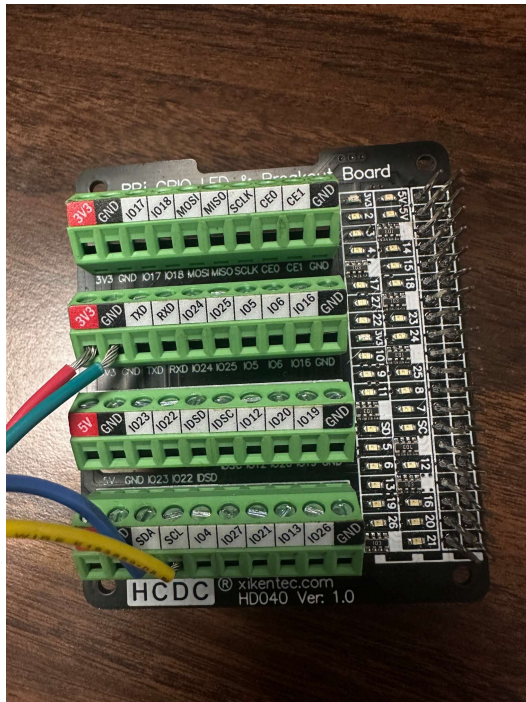


5.3 Wiring

	≡ MPU-6050	≡ Raspberry Pi
1	VCC	3.3V
2	GND	GND
3	SDA	GPIO2
4	SCL	GPIO3

The Raspberry Pi I²C interface must be enabled.





6. Software Design

The application was developed in Python and runs within a dedicated virtual environment.

Required libraries include:

```
1 numpy
2 matplotlib
3 mpu6050-raspberrypi
4
```

Python ▾

The software performs:

1. Data acquisition
2. Signal conditioning
3. FFT analysis
4. Damping estimation
5. Report generation
6. Chart generation

7. Data Acquisition Process

The operator:

1. Starts a measurement
2. Performs a light impact on the tool or holder

3. Allows vibration decay to occur

The program records:

1 Time
2 Acceleration X
3 Acceleration Y
4 Acceleration Z
5

into a timestamped CSV file.

An 8-second capture period was selected to improve damping calculation accuracy and provide sufficient resolution for FFT analysis.

	A	B	C	D	E	F
1	time s	accel x	accel y	accel z	g	
2	0.003	-0.4477	-9.9647	0.0958		
3	0.006	-0.4166	-9.9407	0.146		
4	0.009	-0.4381	-9.9359	0.0527		
5	0.0119	-0.4214	-9.924	0.1197		
6	0.0149	-0.3926	-9.912	0.0168		
7	0.018	-0.4238	-9.8641	0.103		
8	0.021	-0.4501	-9.8857	0.0455		
9	0.0239	-0.4956	-9.9288	0.0886		
10	0.0269	-0.3711	-9.9407	0.1101		
11	0.0299	-0.4166	-9.9886	0.1556		
12	0.0332	-0.3926	-9.9599	0.158		
13	0.0367	-0.431	-9.9048	0.0383		
14	0.0398	-0.4166	-9.9048	0.079		
15	0.0427	-0.4094	-9.8928	0.1365		
16	0.0457	-0.407	-9.8976	0.1221		
17	0.0486	-0.3926	-9.8354	0.2682		
18	0.0516	-0.4429	-9.9216	0.0239		
19	0.0545	-0.3807	-9.9671	0.1053		
20	0.0575	-0.419	-10.1035	0.0311		
21	0.0604	-0.4501	-9.9551	0.1389		
22	0.0634	-0.395	-9.9264	0.1245		
23	0.0663	-0.3926	-9.9551	0.0934		
24	0.0693	-0.4381	-9.9192	0.1101		
25	0.0723	-0.34	-9.9695	0.0838		
26	0.0753	-0.3879	-9.9048	0.0646		
27	0.0783	-0.3759	-9.9599	0.1269		
28	0.0813	-0.4286	-9.9503	0.1221		
29	0.0843	-0.431	-9.9431	0.1006		
30	0.0872	-0.419	-9.9527	0.0455		
31	0.0902	-0.4405	-9.9096	0.2538		
32	0.0931	-0.4357	-9.9671	0.17		
33	0.096	-0.4525	-9.9192	0.0766		
34	0.099	-0.3974	-9.9216	0.1293		

8. Signal Processing

8.1 DC Removal

Before analysis, mean acceleration values are removed.

This centers the signal around zero and prevents low-frequency distortion.

8.2 Windowing

A Hanning window is applied.

Benefits include:

- Reduced spectral leakage
- Improved frequency identification
- Cleaner FFT plots

9. Fast Fourier Transform Analysis

The FFT converts acceleration data from:

1 Time Domain

2

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into

1 Frequency Domain

2

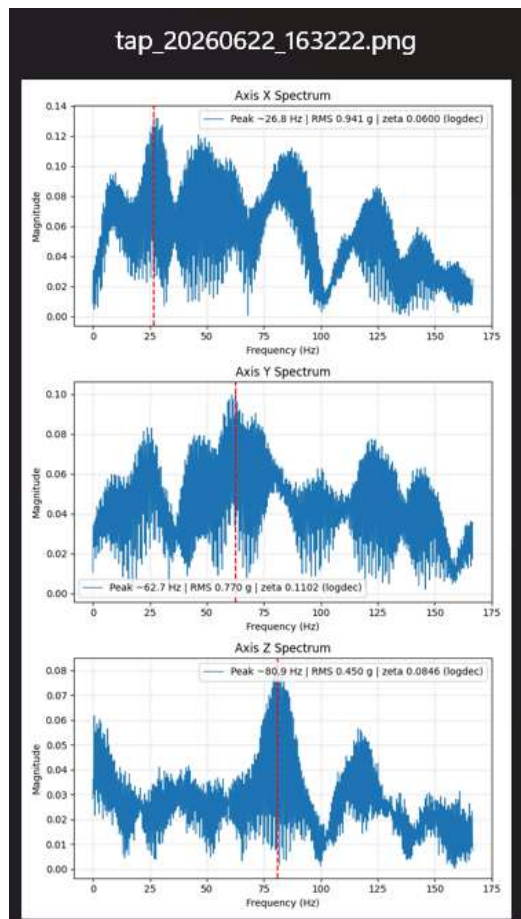
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allowing vibration energy to be visualized as a function of frequency.

The FFT identifies:

- Dominant frequencies
- Relative vibration amplitudes
- Resonant modes

for each axis.



10. RMS Vibration Measurement

Root Mean Square vibration is calculated using

$$RMS = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2}$$

RMS provides a measure of overall vibration of energy.

It is useful for comparing relative vibration severity between axes.

11. Damping Estimation

Two damping estimation techniques are implemented.

11.1 Logarithmic Decrement

The primary damping method uses free-decay vibration peaks.

$$\delta = \frac{1}{n} \ln \left(\frac{x_1}{x_{n+1}} \right)$$

Damping ratio is then calculated as:

$$\zeta = \frac{\delta}{\sqrt{4\pi^2 + \delta^2}}$$

This method is particularly effective for tap-testing applications because it directly analyzes the vibration decay response.

11.2 Half-Power Bandwidth Method

The FFT resonance peak width provides an independent damping estimate.

$$\zeta = \frac{f_2 - f_1}{2f_n}$$

where:

- f_n = resonance frequency
- f_1 = lower half-power frequency
- f_2 = upper half-power frequency

This result is used primarily as validation of the logarithmic decrement method.

12. Chatter Axis Identification

Vibration occurs in all three axes, but not all axes contribute equally to chatter.

The software automatically:

1. Evaluates all detected dominant frequencies
2. Compares spectral energy levels
3. Selects the lowest significant flexible mode

This mode is identified as the primary chatter-driving vibration.

```
File  Edit  View

File: tap_20260622_163222.csv (Duration ~7.999 s, Fs ~333.3 Hz)
Axis X: Peak ~26.8 Hz (RMS 0.941 g, damping zeta 0.0600, method logdec)
Half-power check: f1 ~26.7 Hz, f2 ~26.9 Hz, bandwidth ~0.2 Hz
Axis Y: Peak ~62.7 Hz (RMS 0.770 g, damping zeta 0.1102, method logdec)
Half-power check: f1 ~62.6 Hz, f2 ~62.8 Hz, bandwidth ~0.2 Hz
Axis Z: Peak ~80.9 Hz (RMS 0.450 g, damping zeta 0.0846, method logdec)
Half-power check: f1 ~80.8 Hz, f2 ~81.2 Hz, bandwidth ~0.4 Hz
Primary chatter axis selected: Axis X (lowest significant flexible mode at ~26.8 Hz)
Recommendation: Avoid speeds causing ~27 Hz vibration.
(For a 2-flute tool, ~803 RPM may excite this mode.)
Estimated damping ratio on primary axis: zeta ~0.0600
Avoid-these-speeds chart saved to: tap_20260622_163222_avoid_chart.png
```

13. Spindle Speed Recommendation

The tooth-passing frequency is defined as:

$$TPF = \frac{RPM \times Z}{60}$$

Where:

- RPM = spindle speed
- Z = number of flutes

When tooth-passing frequency coincides with a resonance frequency, chatter may occur.

The software calculates:

$$RPM = \frac{60 \times f_n}{Z}$$

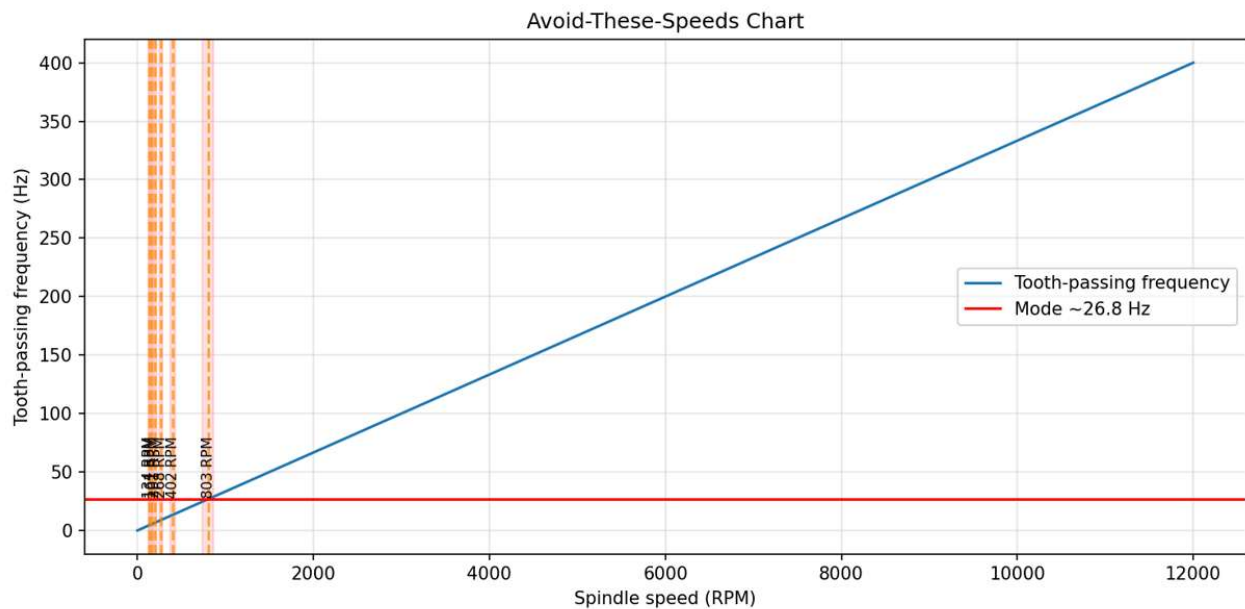
and generates recommended avoidance zones.

14. Automatic RPM Avoidance Charts

The software automatically generates a spindle-speed avoidance chart showing:

- Resonant frequencies
- Tooth-passing frequencies
- Chatter risk zones
- Predicted critical RPM values

These charts are intended to provide practical guidance during machine setup.



15. Validation Results

Initial testing demonstrated successful detection of multiple vibration modes.

Example measurements:

	≡ Axis	≡ Frequency
1	X	26.9 Hz
2	Y	62.8 Hz
3	Z	80.8 Hz

The X-axis mode was automatically selected as the primary chatter mode because it represented the lowest significant flexible resonance.

For a two-flute cutter:

RPM = 803

was identified as a speed likely to excite this vibration mode.

16. Limitations

The current implementation provides practical spindle-speed avoidance recommendations but does not generate a full industrial Stability Lobe Diagram.

A complete stability lobe implementation would require:

- Instrumented impact hammer
- Frequency Response Function (FRF)
- Modal stiffness estimation
- Cutting force coefficient characterization

These capabilities are planned for future development.

17. Future Enhancements

Planned improvements include:

Phase 2

Instrumented hammer support (connecting electronically a small hammer to the RPi to provide input force data for calculation).

Phase 3

Frequency Response Function measurement

Phase 4

Stability lobe chart generation

Phase 5

Web dashboard

Phase 6

Machine learning chatter prediction

Phase 7

Cloud-based tool database (to select the exact tool and its properties for improved analysis)

18. Conclusion

This project demonstrates that a Raspberry Pi 4 and MPU-6050 accelerometer can be used to create a functional CNC vibration analysis platform capable of identifying resonance frequencies, estimating damping ratios, and generating practical spindle-speed recommendations.

The system provides many of the benefits of commercial tap-testing platforms while remaining affordable, portable, and open source.

The resulting platform offers an accessible introduction to modal analysis and machine dynamics while delivering immediate value to CNC machinists, manufacturing engineers, and educators.

References

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